Reproducible Low-Dimensional Structure Can Fail a Causal-State Test in Neural Training

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Abstract

Low-dimensional structure in neural-network training can suggest a compact state description, but reproducibility and prediction do not by themselves establish causal adequacy. We test a pre-specified causal-state criterion in clean ResNet18-GN training with AdamW. Twenty discovery tasks fix the representation and analysis; 30 disjoint tasks are then evaluated untouched. Function-preserving interventions modify only AdamW first-moment memory, leaving model parameters and instantaneous logits unchanged. Two independent query families reconstruct a consistent one-dimensional within-task coordinate (mean correlation 0.777, 95% CI $[0.700, 0.845]$), yet a discovery-frozen decoder predicts the $h = 16$ intervention-caused displacement worse than the zero-change baseline under both held-out query families. A separate confirmation-only control calibrates optimizer-state directions to the same one-step logit effect: the historical AdamW first-moment direction achieves that effect with only $1/90.93$ of the median displacement and $1/70.79$ of the most potent displacement among eight matched history-orthogonal random controls. Thus, within this controlled system, reproducible low-dimensional structure can fail a direct causal-state test, while historical optimizer memory is exceptionally potent relative to a pre-specified orthogonal random-direction null.

Question and causal criterion

Deep-network training can occupy surprisingly low-dimensional regions of prediction space (Mao et al. 2024). A compact representation, however, may summarize observed trajectories without preserving what happens under intervention. Causal abstraction makes this distinction explicit: a useful high-level description should preserve relevant interventional behavior, not only correlational structure (Geiger et al. 2025).

We therefore test one operational criterion. Let s_t be the full trainer state at a checkpoint, let s'_t differ only by a hidden optimizer-state intervention that leaves the current network function unchanged, and let Y_{t+16} be a held-out future response. The caused displacement is

$$\Delta Y_{16} = Y_{t+16}(s'_t) - Y_{t+16}(s_t).$$

For this experiment, a candidate scalar passes the causal-state test only if a decoder frozen on discovery tasks predicts ΔY_{16} on untouched tasks better than the physically correct zero-change predictor. We report causal gain as

$$G = 1 - \frac{\mathrm{MSE}(\widehat{\Delta Y}, \Delta Y)}{\mathrm{MSE}(0, \Delta Y)},$$

so $G < 0$ means the learned predictor is worse than predicting no caused change.

Prospectively frozen experiment

Each task is an independently generated clean four-class synthetic classification problem ($K = 32$ latent dimensions, $r = 4$ task-relevant dimensions). ResNet18-GN models are trained for 500 AdamW updates. Twenty tasks are used for discovery and 30 disjoint tasks for confirmation; task identity, dimension comparison, regularization, decoders, thresholds, and claim definitions are frozen before confirmation outcomes are evaluated.

Each query family is a bank of independently generated, locally calibrated signed parameter perturbations applied at matched target modules. On discovery tasks, regularized CCA identifies a shared low-dimensional coordinate from the Q_A and Q_B short-horizon response vectors at $h \in \{1, 2, 4\}$; a linear decoder fitted only on discovery then predicts the physically distinct Q_T response at $h = 16$. Confirmation applies these frozen transforms without refitting. At the fixed 70% checkpoint, the intervention changes only AdamW's first moment; parameters, buffers, second moments, step counters, and current logits are unchanged. Across all 210 confirmation branches, exact current-function invariance, optimizer-field isolation, and pre-specified locality checks pass. Recent work independently treats optimizer moments as part of the dynamical state of learning (Liu and Li 2026; Xu 2026).

Confirmation results

The one-dimensional coordinate is reproducible across independent measurements: mean within-task correlation is $r = 0.7765$ (95% CI $[0.7002, 0.8449]$), with positive agreement on all 30 confirmation tasks. The one-dimensional representation also remains within the pre-specified tolerance of its discovery-selected higher-dimensional challenger.

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Table 1: Frozen design and primary confirmation endpoints.

Component	Frozen choice
Discovery / confirmation	20 / 30 independent tasks
Coordinate measurements	signed local-query banks Q_A, Q_B, $h = 1, 2, 4$
Held-out future target	disjoint signed-query bank Q_T, $h = 16$
Hidden intervention	AdamW first moment only
Primary causal baseline	$\widehat{\Delta Y} = 0$
Inference unit	task/replicate

Table 2: Primary confirmation results ($n = 30$ tasks). Intervals are task-level bootstrap 95% CIs.

Test	Estimate (95% CI)
Independent-query agreement	$r = 0.777$ $(0.700, 0.845)$
Causal gain vs. zero, Q_A	-0.0218 $(-0.0359, -0.0090)$
Causal gain vs. zero, Q_B	-0.0226 $(-0.0376, -0.0091)$
History / median orthogonal control	$90.93\times$ $(75.95, 94.57)$
History / best of 8 orthogonal controls	$70.79\times$ $(58.96, 82.00)$

The direct causal test nevertheless fails. Using the prespecified gain G above, mean gain relative to predicting no caused change is -0.0218 (95% CI $[-0.0359, -0.0090]$) for Q_A and -0.0226 (95% CI $[-0.0376, -0.0091]$) for independent Q_B. Both intervals lie below zero. The scalar is therefore reproducible without improving prediction of the intervention-caused future displacement.

The same untouched cohort reveals a separate positive result. For this confirmation-only control, each optimizer-state direction is calibrated to produce the same prospective one-step logit RMSE; functional potency is therefore the inverse first-moment displacement required to reach that common effect. Eight Gaussian controls are normalized to the history-axis norm and, by pre-specified design, constructed in the orthogonal complement of history. The median control requires $90.93\times$ the first-moment displacement required by history (95% CI $[75.95, 94.57]$), and even the most potent of the eight controls requires $70.79\times$ as much (95% CI $[58.96, 82.00]$). Every confirmation task exceeds the prespecified task-level potency floor. This result is therefore a comparison to a matched history-orthogonal null, not a claim about unrestricted random directions in the full optimizer-state space.

Why weaker evidence can mislead

The final experiment deliberately separates properties rather than using a master vote. Across tasks, an effectively one-dimensional phenotype predicts absolute held-out future responses better than a rich observational baseline under both independent query families (mean relative improvements 0.114 and 0.105), yet the two query views do not recover the same task ordering ($r = 0.203$, 95% CI $[-0.184, 0.560]$). Within task, the identity problem reverses: independent reconstructions agree strongly ($r = 0.777$), but the scalar loses to the zero-change causal baseline. Thus low dimension, prediction, and independent reproducibility can dissociate under stronger tests.

The zero-change comparator is crucial because the target is a *caused displacement*, not an absolute future state. If an intervention has little predictable downstream effect, zero is the correct counterfactual prediction. Indeed, the within-task scalar beats a richer observational/treatment baseline for centered future effects, yet still loses to zero. The stronger causal comparator therefore prevents a weak observational baseline from creating a misleading success.

Reliability, scope, and interpretation

The independent inferential unit is always the task, not individual response rows. Confidence intervals are task-level bootstrap intervals and directional tests use task-level sign flips. The query families must pass a prospective geometric-independence gate; every hidden-state branch must pass exact current-function invariance and locality checks. All retained confirmation measurements satisfy the same prespecified query-support condition, and the frozen estimators were unchanged throughout confirmation; completion and provenance details are documented in the supplement.

The claim is intentionally narrow. These results concern controlled synthetic clean-learning tasks, one ResNet18-GN family, AdamW, and a specified local intervention family. They do not establish architectural or natural-data universality. They do provide a controlled counterexample to inferring causal statehood from reproducible low-dimensional structure in the tested system. The independent potency result instead suggests a mechanistic hypothesis: historical optimizer memory may align with unusually high-gain control directions even when a compact predictive coordinate is not itself an adequate causal state.

References

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